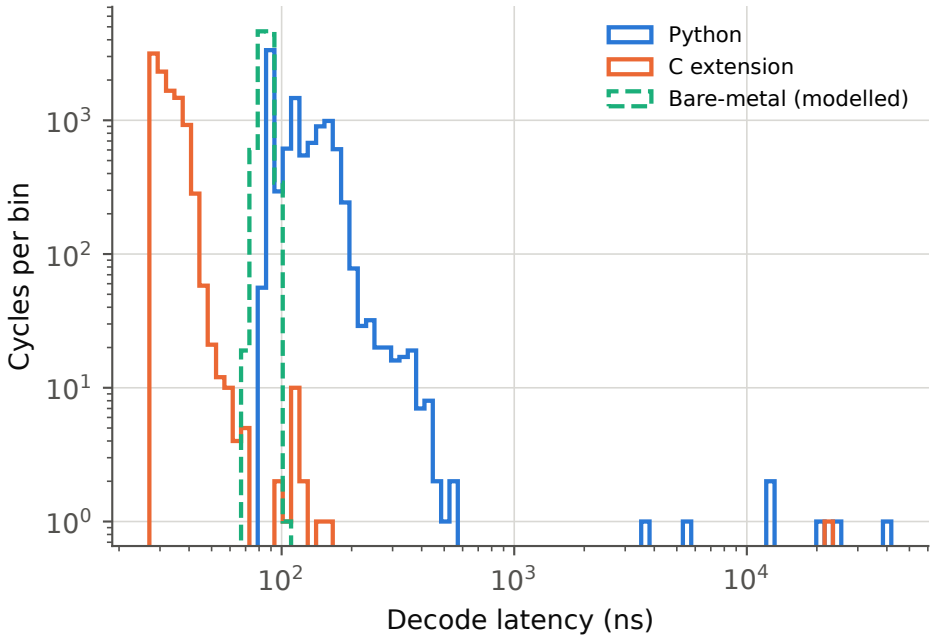
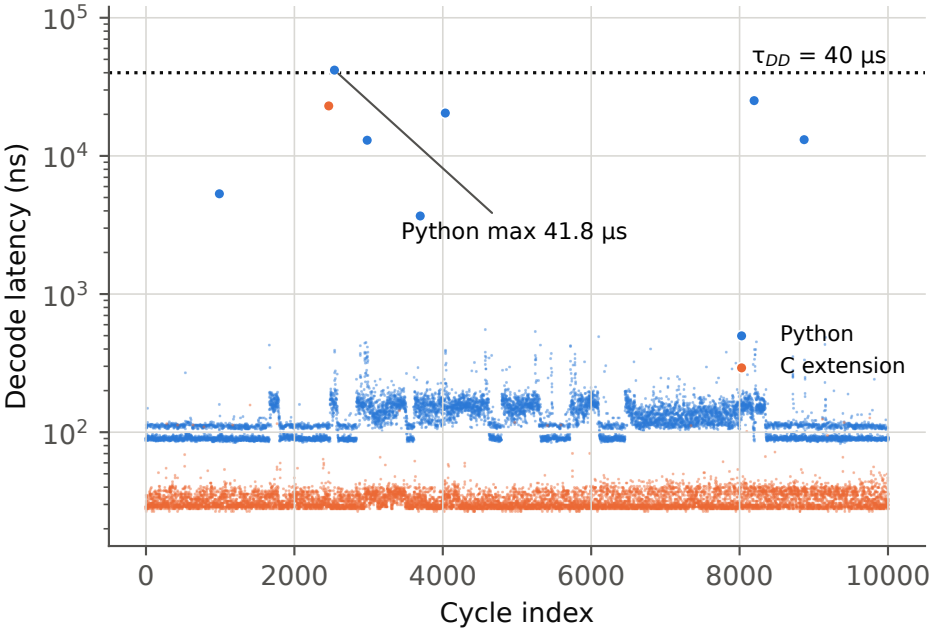
Decode latency, 10,000 cycles per tier — Swayam Shakya

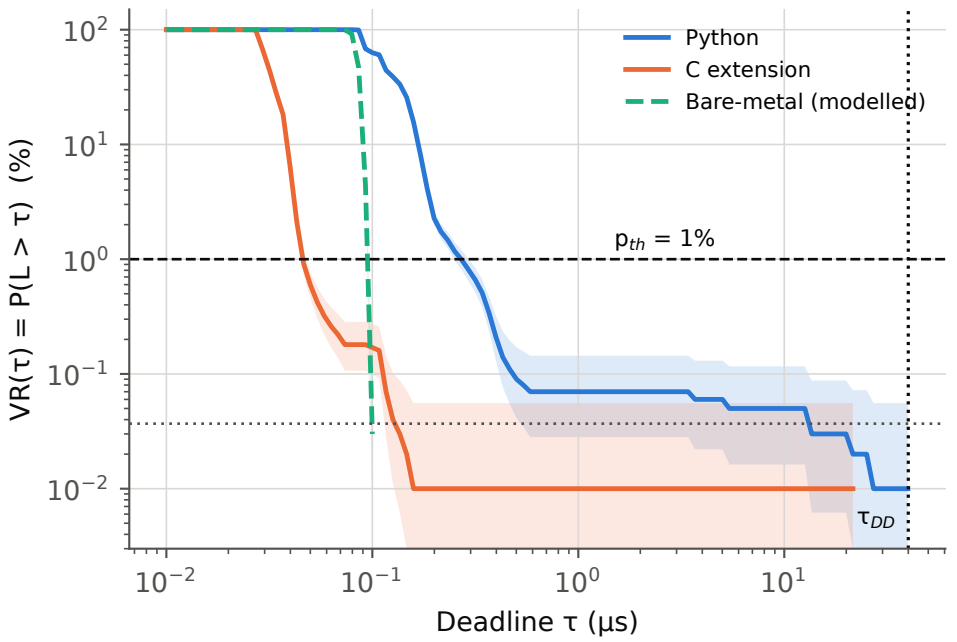
(A) Latency distributions



(B) Jitter: every cycle in order; spikes > 1 µs enlarged



(C) Violation rate; bands = exact 95% CI



(D) Summary (ns unless stated)

Tier	Median	Timer floor	P99	Max	VR@40 µs (%) [95% CI]
Python	113	63	268	41,786	0.01 [0.00, 0.056]
C extension	31	26	46	23,009	0.00 [0.00, 0.037]
Bare-metal*	85	—	95	102	0.00 [0.00, 0.037]

* modelled, not measured: Cortex-M7 at 480 MHz, 41-cycle WCET

Dotted line in (C): 95% upper bound on VR when 0 of 10,000 cycles miss (0.037%)